

FORM TP 2005199



TEST CODE **02120032**

MAY/JUNE 2005

**CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION**

ENVIRONMENTAL SCIENCE

UNIT 1: ECOLOGY, PEOPLE AND NATURAL RESOURCE USE

PAPER 03B

2 hours

31 MAY 2005 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of **THREE** questions.
2. Answer **ALL** questions.
3. Write your answers in the answer booklet provided.
4. Graph paper is provided.
5. The use of a non-programmable calculator is allowed.

Answer ALL questions.

In 1990, a rare species of parakeets was discovered in a remote forest location where development activity or human interference was not evident. This population of parakeets was observed and monitored from 1991 to 2000. The estimated number of parakeets in the population between 1991 and 2000 is indicated in Table 1.

TABLE 1: ESTIMATED NUMBER OF PARAKEETS

Year	Estimated Number
1991	550
1992	630
1993	730
1994	820
1995	950
1996	970
1997	960
1998	930
1999	820
2000	780

In 1996, a developer was granted a concession for the construction of lodges. These lodges were to provide accommodation for the increasing number of tourists and birdwatchers who wished to visit the forest location and observe the parakeets. A part of the area was cleared to construct lodges, viewing galleries and other infrastructure.

To increase the attraction of the area for bird watching, the developer proposed to introduce two new species of birds.

1. (a) Use the information in Table 1 to plot an appropriate graph showing the variation in the size of the population of parakeets from 1991 to 2000. [12 marks]
- (b) With reference to your graph:
 - (i) Describe the variation of the parakeet population between 1991 and 2000. [6 marks]
 - (ii) Suggest plausible explanations for the variation in the parakeet population identified in (b) (i). [10 marks]
- (c) From your graph, estimate the carrying capacity of the ecosystem for parakeets. [2 marks]

Total 30 marks

2. Evaluate the possible impact of EACH of the following actions of the developer on the parakeet population:

(a) The introduction of the new species [10 marks]

(b) Clearing an area of the forest [10 marks]

(c) Presence of the visitors [10 marks]

Total 30 marks

3. Design a monitoring plan to determine the impact of the development on the parakeet population.

Total 30 marks

END OF TEST